MIL-A-45652 (Ord)
30 December 1959

SUPERSEDING
FCS-878
4 June 1956

MILITARY SPECIFICATION

AMPLIFIER, INTERMEDIATE FREQUENCY, AM-1062/MS

1. SCOPE

1.1 This specification covers one type of intermediate frequency amplifier, hereinafter referred to as "amplifier". This amplifier is a component of the assemblies and systems listed in 6.2.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1285

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DRAWINGS

Ordnance Corps

E7621810	Amplifier, Intermediate Frequency AM-1062/MS
E7622067	Amplifier, Intermediate Frequency AM-1062/MS, Schematic Diagram

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one amplifier, representative of production, shall be made available to the Government to determine if the amplifier meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The amplifier shall be manufactured in accordance with Drawing E7621810, "Amplifier, Intermediate Frequency AM-1062/MS" and drawings pertaining thereto. The requirements of this specification are detailed only to extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence capabilities prior to final acceptance by the Government.

3.4 Ground potential.- The design and construction of the amplifier shall insure that all external parts are at ground potential at all times.

3.5 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish painting and protective
(e) Electron tubes
(f) Workmanship

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3.6 Environmental.-

3.6.1 Vibration.- The amplifier shall be subjected to a simple harmonic motion having an amplitude of 0.03-inch (0.06-inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the amplifier shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.6.2 Operating temperatures.- The amplifier shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.6.3 Storage temperatures.- The amplifier shall show no evidence of physical failure and shall meet all performance requirements of this specification at the standard temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit).

3.7 Electrical.

3.7.1 Continuity.- The amplifier shall demonstrate continuity in accordance with Electrical Schematic Drawing E7622067.

3.7.2 Dielectric strength.- There shall be no puncture of insulated areas when the amplifiers are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground,and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current applied.

3.8 Performance.- When set-up in accordance with pretest information, specified in Section 4, the amplifier shall meet the following requirements (all symbols referred to are those shown on Drawing E7621810).

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3.8.1 Bandwidth and center frequency.

3.8.1.1 Bandwidth.- The bandwidth measured 3 decibles down from the maximum response shall not be less than 4 megacycles.

3.8.1.2 Center frequency.- The center frequency of the mission band between the 3 decibel points shall be 60 megacycles plus or minus 1.5 megacycles.

3.8.2 Gain.- The maximum gain of the amplifier shall be at least 100 microamperes.

3.8.3 Overload.

3.8.3.1 When a 5 microsecond pulse, as shown in Figure 1 (a), at a minimum level of 5 decibels above the overload point is applied to the amplifier, the output pulse shall be as shown in Figure 1(c).

3.8.3.2 Utilizing the set-up in 3.8.3.1 and terminating an oscilloscope probe as shown in Figure 1(b) the output pulse shall be as shown in Figure 1(d).

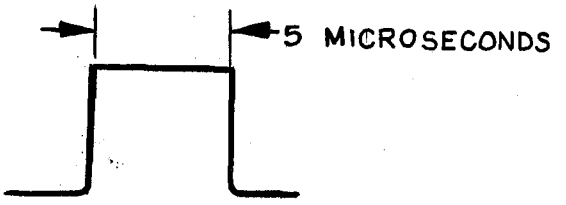
3.9 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

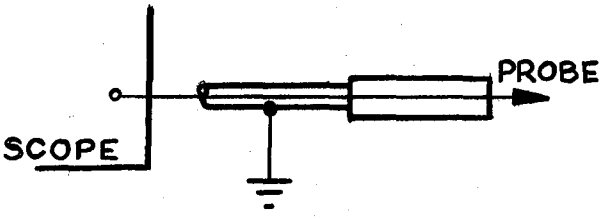
4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Inspection of components.- Contractor's inspection for all components shall be in accordance with the inspection procedures of the written inspection plan required by 4.2 and 4.2.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements that have a bearing on the functional accuracy and performance of the unit.

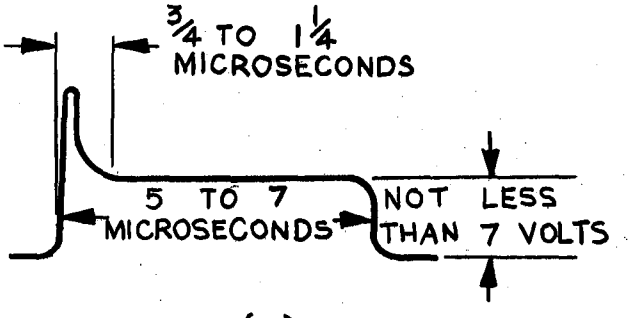
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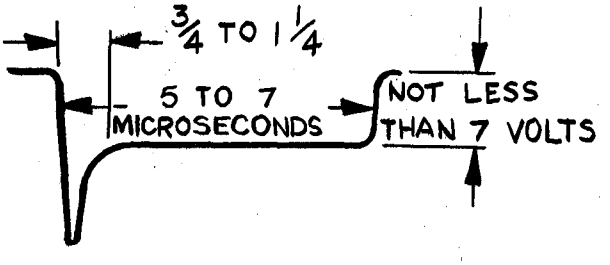
(a)



(b)



(c)



(d)

FIGURE 1

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4.1.1.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.1.1.2 Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection of the characteristics that caused the lot to be rejected.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

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4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.2.1.1 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Classification of tests.- The inspection and testing of the amplifier shall be classified as follows:

(a) Production test.
(b) Acceptance test.

4.4 Production model.

4.4.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.4.2 These tests shall be conducted at the contractor's facility using test equipment designed or procured for use on the remaining items.

4.4.3 The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

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4.4.3.1 Disposition of rejected production model.- Should the production model fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the production model. When corrections in the process have been made, a new production model shall be completely inspected by the Government. The rejected production model shall be corrected, where practicable, and shall become part of the first production lot.

4.5 Inspection provisions.

4.5.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratorie The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.5.2 Acceptance testing.- Sampling of the amplifier for acceptance may be used in lieu of 100 percent inspection. The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items. Samples selected for acceptance testing shall include the samples selected for environmental testing.

4.5.3 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.4 Disposition of rejected lots. Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L6, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL value is expressed in precent defective.

MAJOR:	AQL 4.0%	Req't Par.	QA Par.	Test Method
1.	Physical failure (vibration)	3.6.1	4.8.1.2	Visual
2.	Bandwidth (-40°F, +150°F)	3.8.1.1	4.8.1.3	SME
3.	Center frequency (-40°F, +150°F)	3.8.1.2	4.8.1.3	SME
4.	Gain (-40°F, +150°F)	3.8.2	4.8.1.3	SME
5.	Overload (-40°F, +150°F)	3.8.3	4.8.1.3	SME

TABLE III - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL value is expressed in percent defective.

MAJOR:	AQL 1.5%	Req't Par.	QA Par.	Test Method
1.	Continuity	3.7.1	4.8.2.1	SME
2.	Dielectric strength	3.7.2	4.8.2.2	SME
3.	Insulation resistance	3.7.3	4.8.2.3	SME
4.	Bandwidth	3.8.1.1	4.8.3.2	SME
5.	Center frequency	3.8.1.2	4.8.3.2	SME
6.	Gain	3.8.2	4.8.3.3	SME
7.	Overload	3.8.3	4.8.3.4	SME
8.	Identification marking	3.9	4.9	Visual

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4.6 Inspection equipment.

4.6.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance calibration and rehabilitation on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.6.2 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7 Test equipment accuracy requirements.

4.7.1 In addition to the above accuracy requirements specified in 4.6.2, the following test equipment shall be used in conjunction with Figure 2:

Item Number	Description
1.	R.F. signal generator, continuously variable from 50 to 70 megacycles, with completely shielded coaxial output connection, and output impedance of 75 ± 15 ohms.
2.	Microameter shall measure 0 to 200 microamperes, fitted with shielded leads, negative lead grounded, positive lead in series with 10,000 ohms ± 1 percent 1/2 watt resistor.

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Item Number (cont'd)	Description
3.	Pulse generator shall be per ordnance drawing A-7621328 or equal.
4.	Oscilliscope shall be a TS-239/UP or equivalent.
5.	Resistor, 2500 ohms, 1 watt, ± 1 percent.
6.	Attenuator, 60 megacycles.
7.	Resistor, 2500 ohms, 1/2 watt, ± 5 percent.
8.	Capacitor, 24 micromicrofarads, 500 volts, ± 2 percent.
9.	Power supply + 180 volts, ± 1 volt 0.04 ampere min.
10.	Power supply + 150 volts dc, ± 1 volt, 0.05 ampere min.
11.	Power supply, 6.3 volts, 400 cycles, 1.6 amperes.

4.7.2 Device, to test for vibration requirement.- The device shall consist of a vibrating machine which will produce motion in each of the 3 mutually perpendicular planes at a uniformly varied frequency between the limits of 10 and 55 cycles per second and return to 10 cycles. It shall have an amplitude of 0.03 inch (0.06 inch maximum excursion) and shall be equipped with a medium capable of holding the unit securely to the vibrating machine.

4.8 Test methods and procedures.

4.8.1 Environmental.

4.8.1.1 Vibration.- Each amplifier in the sample shall be mounted in its normal operating position on a special testing device and vibrated in accordance with the requirement specified.

4.8.1.2 Physical failure post vibration.- Subsequent to vibration, the amplifier shall be subjected to a visual and tactile examination for damage. The unit then shall be subjected to the tests specified in 4.8.1.3 and 4.8.1.4 inclusive, and shall meet the requirements specified.

4.8.1.3 Operating temperatures.- When thermal equilibrium of the amplifier is reached at plus 150 degrees Fahrenheit and minus 40 degrees Fahrenheit, respectively, the amplifier shall be subjected to the tests specified in 4.8.3, and shall meet the requirements specified.

4.8.1.4 Storage temperatures.- This test shall be conducted in accordance with Procedure II of Specification MIL-F-13926 to the conditions specified. Upon completion of the specified exposure, the unit shall be inspected for damage and then subjected to the test specified in 4.8.3, and shall meet the requirements specified. This test shall be conducted on the production model only.

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4.8.2 Electrical.

4.8.2.1 Continuity.- A continuity check of the amplifier circuits shall be made using standard measuring equipment to determine conformance with the requirements specified.

4.8.2.2 Dielectric strength.- The amplifier shall be tested in accordance with method 301 of Standard MIL-STD-202, and shall meet the requirements specified.

4.8.2.3 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the amplifier shall be given the insulation resistance test in accordance with method 302, test condition B, of Standard MIL-STD-202, and shall meet the requirements specified.

4.8.3 Performance.

4.8.3.1 Pretest information.

4.8.3.1.1 Randomly selected tubes shall be inserted in the proper sockets and the unit shall be installed in its bottom cover. The bottom cover shall be in place with all screws and coaxial cables drawn up tightly during the entire testing procedure.

4.8.3.1.2 The suggested test circuit, figure 2, shall be connected to the amplifier to be tested.

4.8.3.2 Bandwidth and center frequency.

4.8.3.2.1 Bandwidth.- With the signal generator connected to J1, as shown in figure 2, the bandwidth when measured the specified decibels down from the maximum response shall not be less than the specified limit.

4.8.3.2.2 Center frequency.- The center frequency of the transmission band when measured between the 3 decibel points shall be as specified in 3.8.1.

4.8.3.3 Gain.- The microameter shall produce the specified reading in microamperes when the maximum gain of the amplifier is a continuous wave input power of not more than 70 decibels below 2 milliwatt.

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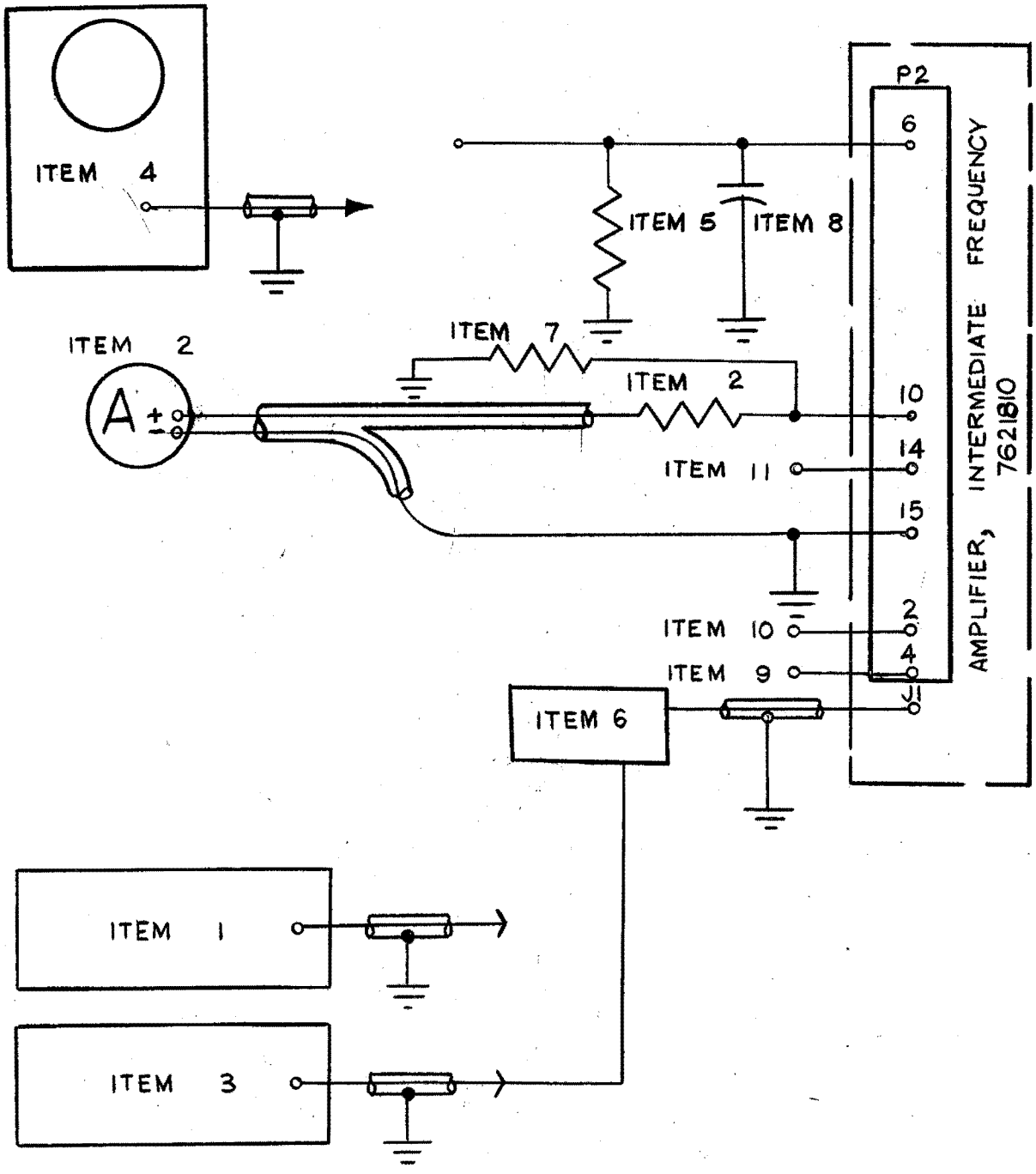


Figure 2

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4.8.3.4 Overload.- With the pulse generator connected to J1 and set for a 5 microsecond pulse, as shown on figure 1(a) at a level above the overload point as specified, the output pulse, as observed at pin 10 of P2, shall closely resemble the waveform shown in figure 1(c) in shape and amplitude. With the oscilloscope probe terminated, as shown in figure 1(b) the output at pin 6 of P2 shall closely resemble the wave form as shown in figure 1(d).

4.9 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The amplifier shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 P1816. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The amplifier receives 60 megacycle IF signals from the Intermediate Frequency Amplifier AM-1059/MS, amplifies them and converts them to video signals for use in the carrier generator.

6.2 Application.- The amplifier is a component of the following assemblies and systems:

Radar Set Group	Antiaircraft Fire Control System M33C (90MM) and M33D (120MM)
Director Station Group OA/711/MSA-7	Radar Course Directing Central AN/MSW-1 & AN/MSW-3
Director Station Group OA/1480/MSA-19	Antiaircraft Guided Missile System Nike Hercules
Control-Power Supply Group	Guided Missile Fire Control System AN/MSW-3A

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6.3 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1)

6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.5 Design information.- Information regarding specific design not covered or referenced by this specification may be obtain from the procuring activity upon the specific request of the contrac

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodian:
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Preparing Activity:
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